

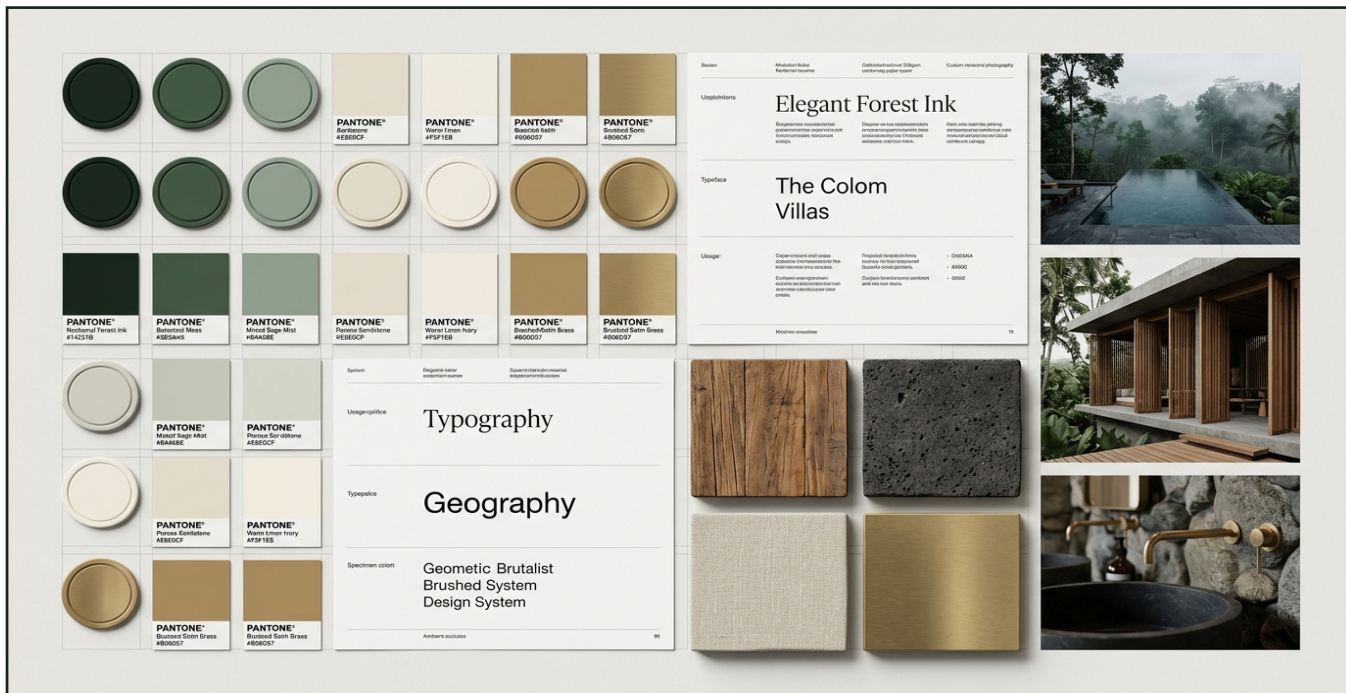
THE TARU VILLAS x VillaOS

TECHNICAL ARCHITECTURE & LUXURY BRAND DESIGN SYSTEM

Direct Booking Engine | Property Management System (PMS) | Ubud, Bali

BRAND DESIGN SYSTEM & AESTHETIC MOODBOARD

Tactile Flatlay: Swiss Grid, Chromatic Specs, PBR Shaders & Architectural Insets



PROJECT IDENTITY

Product: The Taru Villas (Direct Booking + PMS)

Location: Ayung River Valley, Ubud, Bali

Inventory: 12 Luxury Private Pool Villa Units

Core Mission:

Direct guest acquisition with zero OTA commissions + fully autonomous PMS workspace for hotel operations.

Visual Style:

Tropical Brutalism meets Organic Minimalism, calm forest palettes, and brass accents.

Release: v0.1.0-alpha (Opening December 2026)

CORE CAPABILITY PILLARS

1. Real-Time Booking Engine

Server-side rates, dynamic quote, QRIS checkout

2. Double-Allocation Prevention

Database-level room + stay_date unique constraint

3. VillaOS Operations Desk

Arrivals, departures, room assignments, walk-ins

4. Comprehensive Control Center

Daily BAR rates, restrictions, folios, night audit

5. Security & Governance

Role-based authorization & immutable audit logs

6. Production Scalability

Dual SQLite / PostgreSQL migration pipeline

SECTION 1: FULL-STACK TECH STACK ARCHITECTURE

Complete breakdown of core runtime, frontend, database, payments, and security.

CORE RUNTIME & FRAMEWORK

Next.js 16.2.12 (App Router)

React Server Components, Server Actions & API Routes

React 19.2.4 + React DOM

Concurrent rendering & modern hook primitives

TypeScript 5.x

End-to-end static typing across UI & server models

Node.js 20+ Runtime

High-throughput asynchronous I/O backend execution

FRONTEND, STYLING & ANIMATION

Tailwind CSS v4 (@tailwindcss/postcss)

Ultra-fast CSS engine with modern theme tokens

Framer Motion 12.40

Silky hero Ken Burns, marquee, and drawer transitions

react-qr-code v2.2.0

Dynamic real-time QRIS generation on checkout

Next/Font Integration

Sub-resource zero-layout-shift font optimization

DATABASE LAYER & ORM

Prisma ORM 6.19.3

Type-safe database client, migrations, & introspections

SQLite Local Profile (dev.db)

Self-contained zero-config local development database

PostgreSQL Production Target

Automated schema converter & migration deployment

Room-Night Unique Invariant

@@unique([roomId, stayDate]) prevents double-booking

PAYMENT GATEWAY INTEGRATIONS

Xendit Node SDK (xendit-node v7)

Xendit Invoice and QRIS Payment Requests API

Midtrans Payment Gateway

Snap modal checkout & Core API webhook handlers

Mock Payment Sandbox

Offline developer simulator for testing transactions

Cryptographic Webhooks

Token/signature verification & payload hash storage

SECURITY, ROLE-BASED ACCESS CONTROL (RBAC) & COMPLIANCE

ROLE MATRIX (SERVER-ENFORCED)

OWNER	Unrestricted master access: staff management, rates, financial reconciliation, cutover.
MANAGER	Daily operations, rate overrides, folios, reports, maintenance dispatch.
FRONT_DESK	Arrivals, departures, check-in/out, room assignments, walk-in reservations.
HOUSEKEEPING	Room cleaning status (Clean, Dirty, Inspected), maintenance work order requests.
FINANCE	Guest folio billing, manual payments, invoice issuance, refunds & settlements.
VIEWER	Read-only access for audits, management inspection, and operational monitoring.

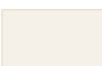
DEFENSE-IN-DEPTH SECURITY PROTOCOLS

- * Strict Content-Security-Policy (CSP) with nonce-less whitelisted origins in next.config.ts
- * Idempotency-Key request headers on all booking & payment mutation endpoints to prevent replay
- * Immutable Operational Audit Trail logging staff actions with timestamp, user ID, and diffs
- * Rate-limited booking endpoints and automatic account lockout on consecutive authentication failures

SECTION 2: BRAND IDENTITY & DESIGN SYSTEM

Colorimetry, typographic anatomy, tactile materials, and spatial guidelines.

CHROMATIC COLOR PALETTE & COLORIMETRY SPECIFICATIONS

	Nocturnal Forest Ink HEX: #14231B PANTONE: Black Forest 19-5408	Deep grounding shade for typography & high contrast
	Botanical Moss HEX: #3D5A45 PANTONE: Forest Biome 19-0315	Primary brand organic green, buttons & active states
	Muted Sage Mist HEX: #8AA08E PANTONE: Muted Sage 15-6304	Secondary botanical tone, borders & sub-labels
	Warm Linen Ivory HEX: #F5F1E8 PANTONE: Gardenia 11-0604	Core canvas background, evoking raw organic linen
	Porous Sandstone HEX: #E8E0CF PANTONE: Sandstone 13-0905	Neutral divider lines, card strokes & table headers
	Brushed Satin Brass HEX: #B08D57 PANTONE: Rich Gold Metallic	Luxury accents, active badges, highlights & rules

TYPOGRAPHIC SYSTEM & HIERARCHY

PRIMARY DISPLAY TYPEFACE (SERIF)

Playfair Display -- Neoclassical Transitional Serif

High stroke-contrast ratio (1:8), acute hairline serifs, ball terminals, tight optical kerning (-15 tracking).

Usage: Page Hero Headings, Section Titles, Villa Names, Luxury Pullquotes.

SECONDARY BODY & UI TYPEFACE (SANS-SERIF)

Jost -- Geometric Neo-Grotesque Sans

Tall x-height, monoline uniform stroke geometry, generous tracking (+40), tabular numeral lining.

Usage: Body Copy, Navigation Bars, Operator Data Tables, Invoices, Financial Folios.

TACTILE MATERIALS & ARCHITECTURAL PALETTE

- Reclaimed Indonesian Teak:	Straight timber grain, satin oil finish, warm golden undertone. (Exterior louvers, custom bed frames, ceiling slats.)
- Honed Volcanic Basalt Stone:	Dark porous volcanic rock, high thermal mass, tactile rough texture. (Private plunge pools, ensuite bathroom feature.)
- Heavy Belgian Linen:	Raw unbleached fiber weave, tactile grain, soft oatmeal hue. (Bedding, drapery, lounge upholstery.)
- Brushed Architectural Brass:	Directional satin brush strokes, subtle natural antique patina. (Plumbing fixtures, door handles, ambient light sconces.)

SECTION 3: OPERATIONS & DEPLOYMENT RUNBOOK

Operator surfaces, verification scripts, backups, and local execution instructions.

VILLAOS PMS OPERATOR SURFACES

<code>/admin</code>	Front Desk Desk
Arrivals, departures, modify/cancel/no-show, check-in/out, room board, housekeeping, walk-in reservations.	
<code>/admin/control -> Rates</code>	Revenue Management
BAR daily pricing, minimum stay rules, Closed to Arrival (CTA), Closed to Departure (CTD), stop-sell.	
<code>/admin/control -> Finance</code>	Guest Accounting
Folio charges, manual payment posting, refunds, balances, printable branded invoices.	
<code>/admin/control -> Maintenance</code>	Facility Dispatch
Work orders, technician assignment, priority, automated out-of-order room blocking.	
<code>/admin/control -> Night Audit</code>	Daily Reconciliation
Close business day, exceptions review, automated daily occupancy & revenue snapshots.	
<code>/admin/control -> Reports</code>	Management Analytics
Occupancy pace, revenue mix, source attribution, downloadable CSV exports.	
<code>/admin/control -> Team</code>	Staff & Security
Staff accounts, role enforcement, session revocation, credential resets.	

OPERATIONAL CLI COMMANDS

<code>npm run dev</code>
Launch local dev server on <code>http://localhost:3000</code>
<code>npm run db:setup</code>
Generate Prisma client, push SQLite schema, seed data
<code>npm run admin:bootstrap</code>
Create secure random local owner account in <code>outputs/</code>
<code>npm run backup</code>
Timestamped SQLite backup with SHA-256 validation
<code>npm run verify</code>
Audit business invariants & room allocation integrity
<code>npm run db:postgres:migrate</code>
Deploy checked-in PostgreSQL migrations
<code>npm run build</code>
Compile optimized production server bundle

AUTHENTICATION & ACCESS POINTS

DEFAULT LOCAL ACCESS:	
URL:	<code>http://localhost:3000/admin</code>
Email:	<code>admin@thetaru villas.local</code>
Password:	<code>Taru-E2zL3qWYsjFnuads-7</code>

SECURITY BEST PRACTICES:	
1.	Change default credentials upon first login.
2.	Create named staff accounts per shift (never share).
3.	Keep <code>outputs/VillaOS_LOCAL_LOGIN.txt</code> confidential.
4.	Set <code>PAYMENT_PROVIDER</code> to <code>xendit</code> or <code>midtrans</code> in prod.
5.	Regularly execute <code>'npm run backup'</code> before audits.
6.	In production, set <code>SEED_DEMO_DATA='false'</code> .

THE TARU VILLAS - VILLAOS 2026
Luxury Architecture x Autonomous Operations